A taste of Competitive Mathematics Doodles

$f(m, n) = ?$

$f(m, n) \leq mn$

Suppose some $p_i \mid q$

(there is some) (p_i divides q)

So $\exists\ k \in \mathbb{Z}$ such that

$$k p_i = q \qquad 1 \le i \le n$$

$$k p_i = m p_i + 1$$

$$m = p_1 p_2 \cdots p_{i-1} p_{i+1} \cdots p_n$$

$$p_i (k - m) = 1$$

Contr bc $\Rightarrow p_i = 1$

$$q \pmod{p_i} = 0 + 1$$
$$= 1$$

$$p_i \nmid q$$

$n = 3$

$a_1 = 2$

$a_2 = 3$

$a_3 = 2$

If $\neg P$

then $a_i - i$ is odd

$\forall i \in \{1, \dots, n\}$

$a_1 - 1$ odd

$a_3 - 3$ odd

$\vdots$

$a_2 - 2$ odd

$a_4 - 4$ odd

$\vdots$

a_1 is even

a_3 is even

$\vdots$

a_2 is odd

a_4 is odd

$\vdots$

$1, 3, 5, \dots, n+1$ even

$2, 4, \dots, n$ odd

$\frac{n-1}{2}$ odd numbers

$\frac{n+1}{2}$ even number

Say $n = 5$

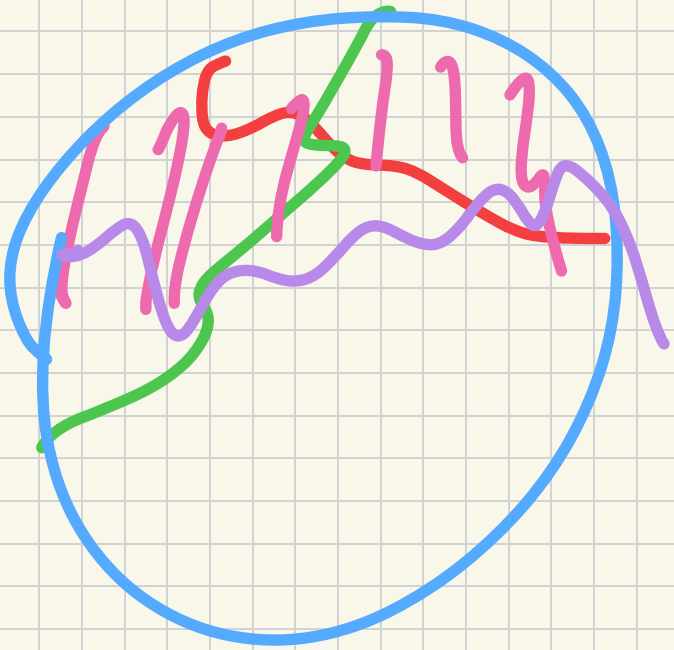
a_1, a_3, a_5 even

a_2, a_4 odd

$[5] = 1, 2, 3, 4, 5$

$n = \frac{9+1}{2} = 5$

$1, 3, 5, 2, 4$ $\quad$ 5



$m > \frac{12}{9}$

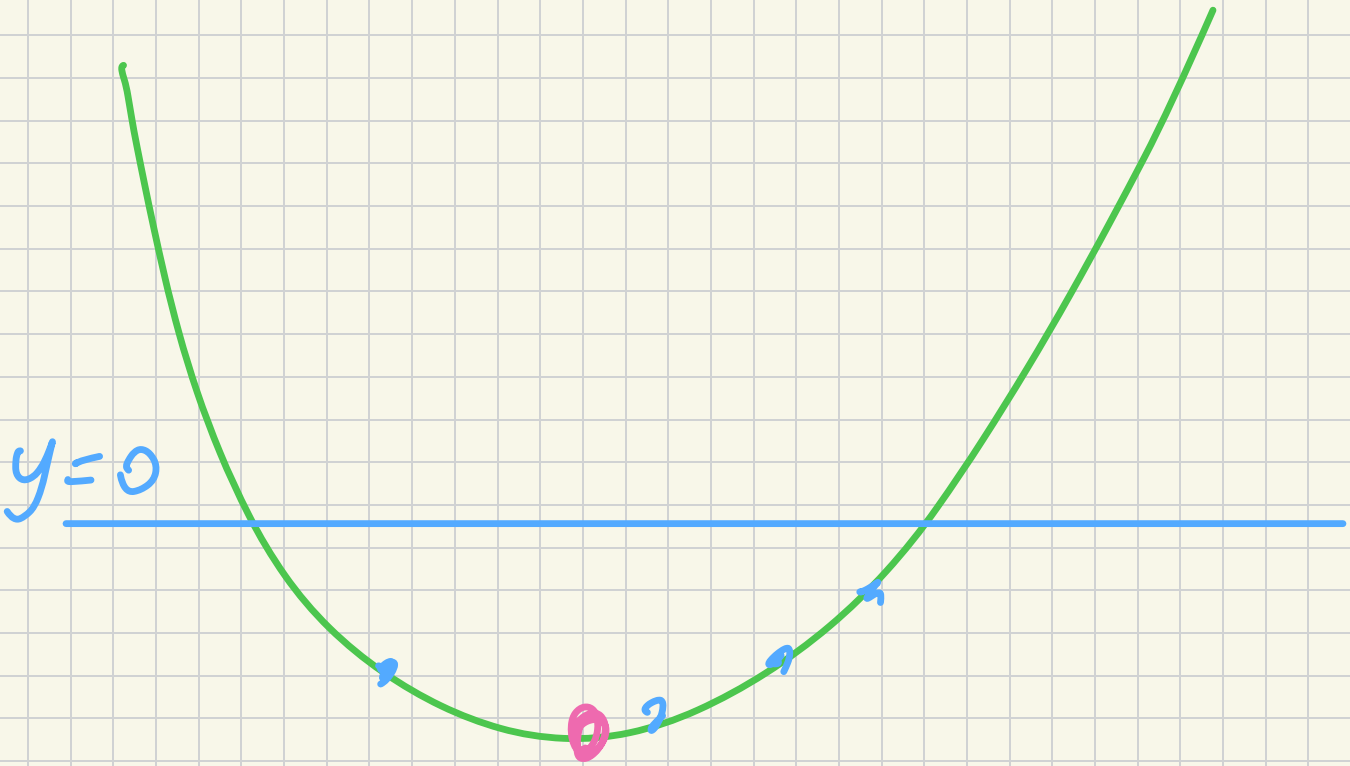
S_1 1

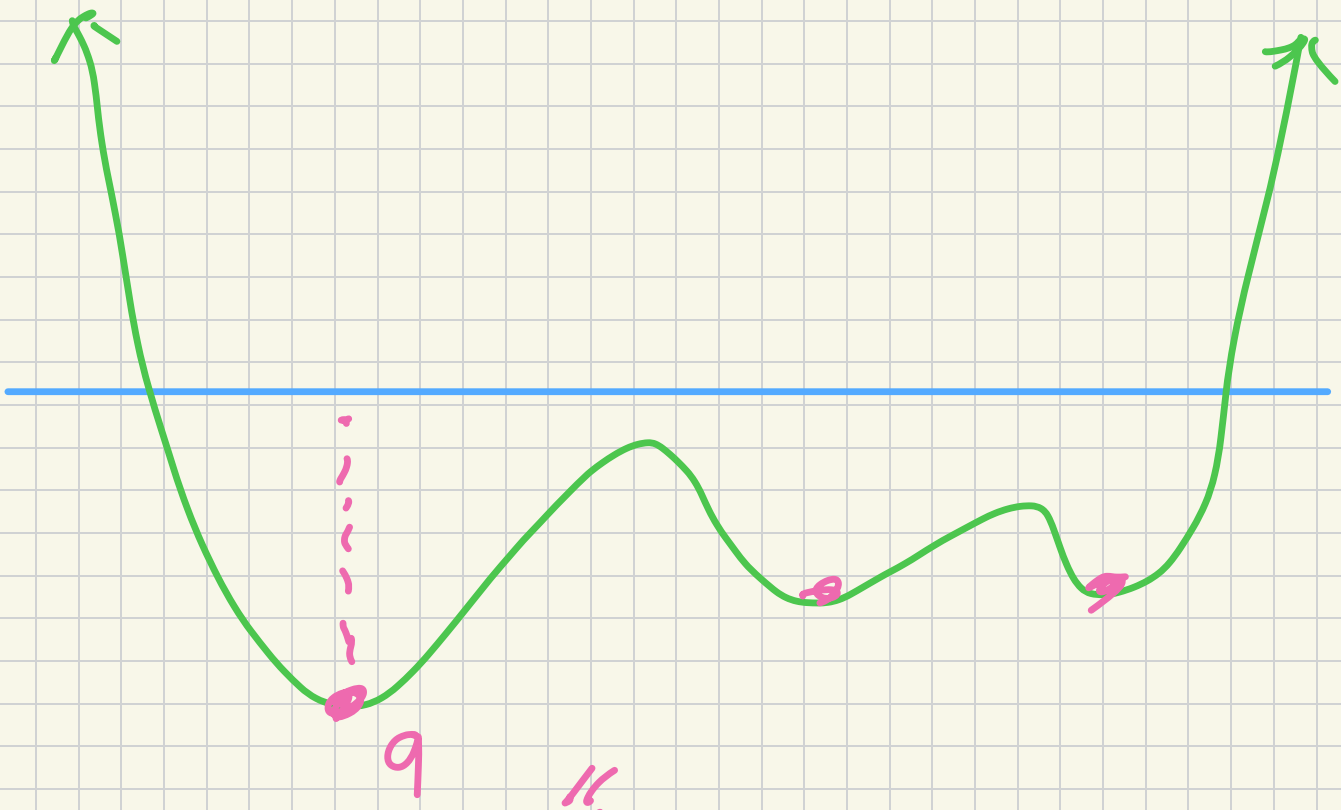
S_2 $\frac{8}{9}$

S_3 $\frac{7}{9}$

$S_1 \cup S_2 \cup S_3$

$1 + \frac{8}{9} + \frac{7}{9}$

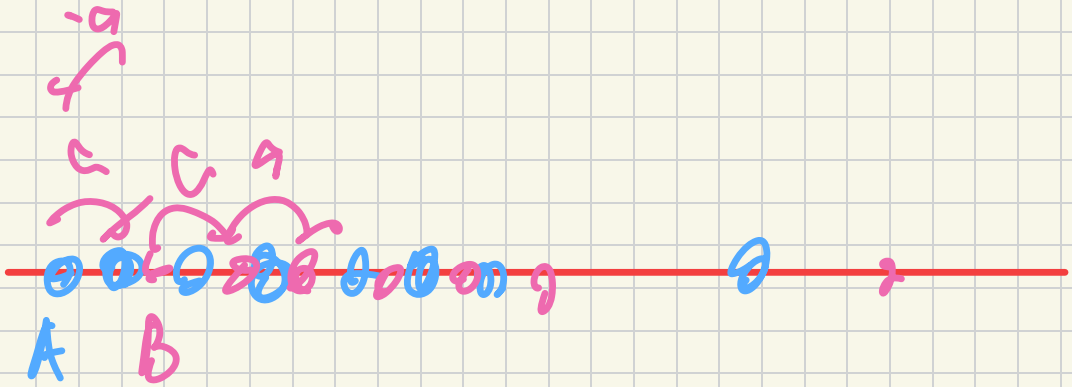




$$p''(a) \geq 0$$

$$P(a) \geq P''(a)$$

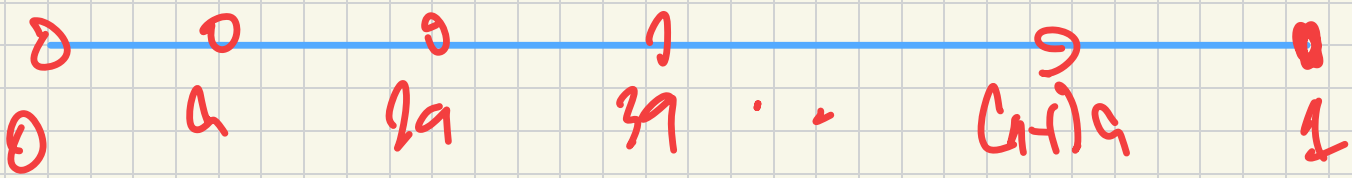
$$P(a) \geq 0$$



$$B = \left\{ x+a \mid x \in A \right\}$$

$$a > 0 \quad x \geq 0$$

If $b \in B$, $b \geq a$.

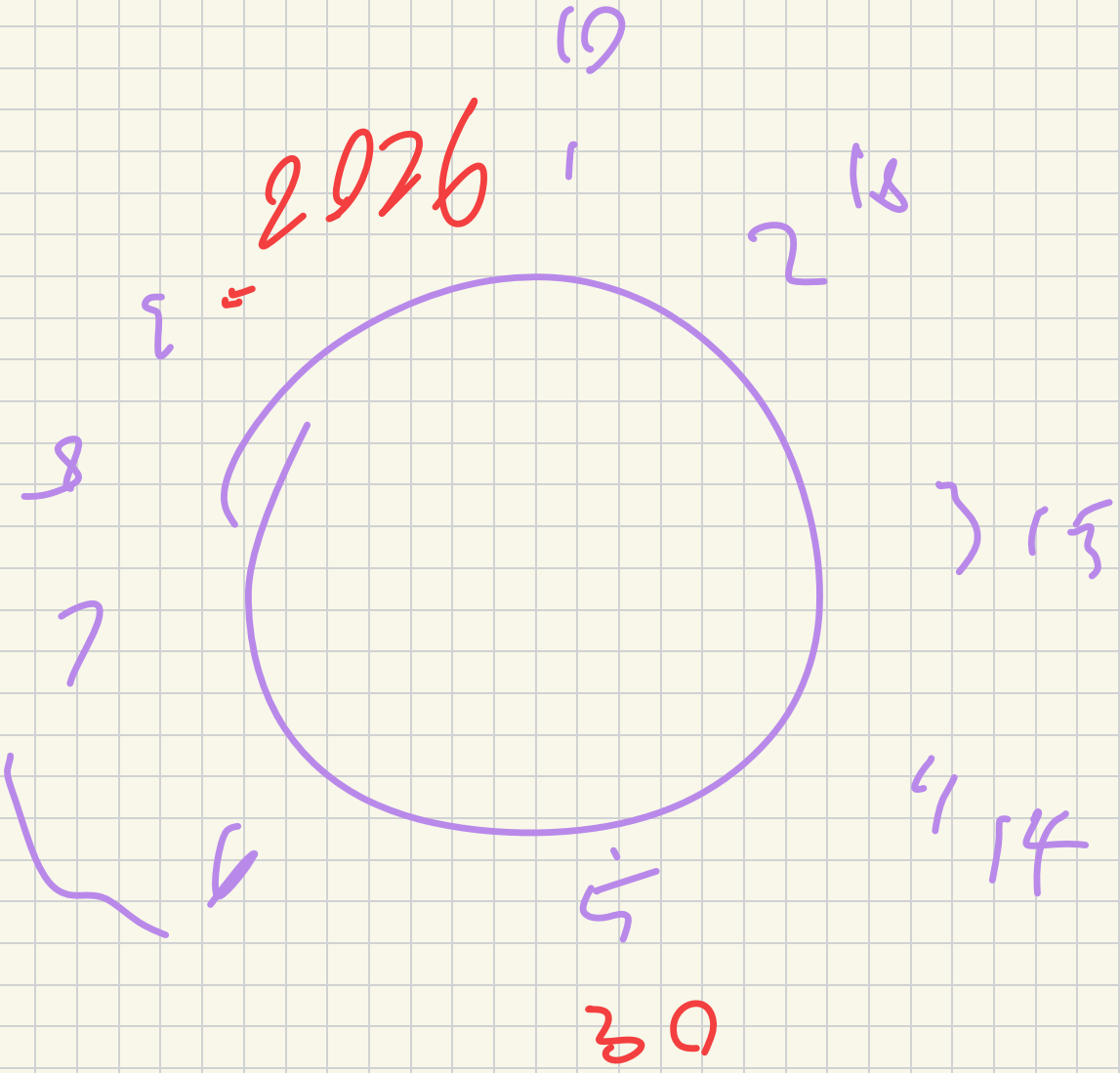


$$\cancel{5} - r^2 \cancel{5} = -2r\sqrt{5} - 2\sqrt{6}$$

$$\frac{r^2}{2} = r\sqrt{5} + \sqrt{6}$$

$$\underbrace{6 + 5r^2}_{\text{Rational}} + \underbrace{2r\sqrt{30}}_{\text{rational}}$$

$$2r\sqrt{30}$$

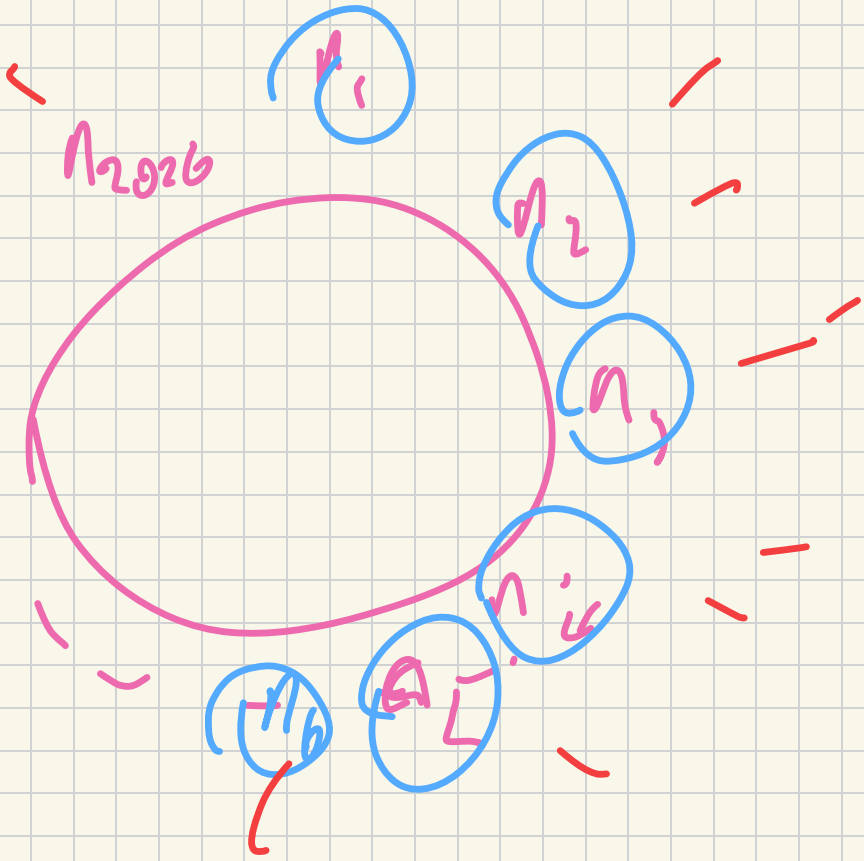


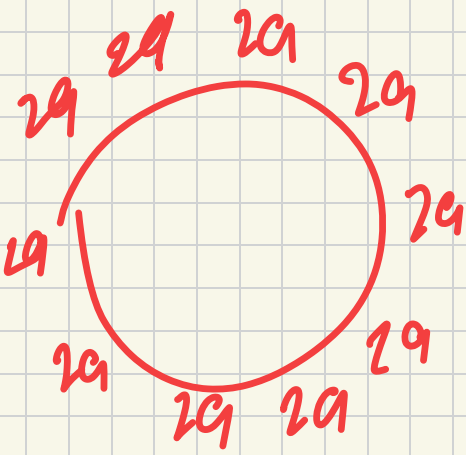
$10-15-18-14+20$

$=21.$

$a - b + c - d + e = 29$

$n_1 - n_2 + n_3 - n_4 + n_5 = 29$





$n_1 - 29$

$n_2 - 29$

$n_3 - 29$

$n_4 - 29$

$n_5 - 29$

$\vdots$

$n_{2026} - 29$

$(n_1, n_2, n_3, n_4, n_5)$

$n_1 - n_2 + n_3 - n_4 + n_5 = 29$

$(n_1 - 29) - (n_2 - 29) \cdots = 29$

$n_1 - \cancel{n_2} + \cancel{n_3} - \cancel{n_4} + \cancel{n_5} = 0.$

$\cancel{n_2} - \cancel{n_3} + \cancel{n_4} - \cancel{n_5} + n_6 = 0$

$n_1 + n_6 = 0.$

n_1, n_6 have same parity.

n_6, n_{11} help same parity.

n_1, n_{16} — —

n_{16}, n_{21} ——

$\downarrow$ ——

$$\begin{array}{ccccc} 2A & 2B & 2C & 2D & 2E \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \end{array}$$

$$a - b + c - d + e = 0$$

$$2A - 2B + 2C - 2D + 2E = 0$$

$$A - B + C - D + E = 0$$